MILITARY SPECIFICATION SHEET

BIT, SCREWDRIVER, TORQ-SET, RIBBED, OFFSET CRUCIFORM, LONG SHANK, EXTERNAL HEXAGON SHANK

This specification sheet is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the bits described herein shall consist of this specification sheet and the latest issue of MIL-B-85643.

1. REQUIREMENTS:

1.1 Material: UNS T41901 or UNS T41902 tool steel in accordance with QQ-T-570, Type S1 or Type S2.

1.2 Dimensions and configuration:

Bit and retaining groove: In accordance with Figure 1 and Table I. Length in accordance with Table II.

Driver point: in accordance with MS14191(AS).

1.2.1 Offset cruciform driver bit gage: In accordance with Table III and Figure 2.

1.3 Hardness: Rockwell 58 to 62 HRC.

1.4 Identification: Stamp military part number (see 1.8) and manufacturer's name, trademark or symbols.

1.5 Test recess: The test recess configuration and dimensions shall conform to Table IV and Figure 3.

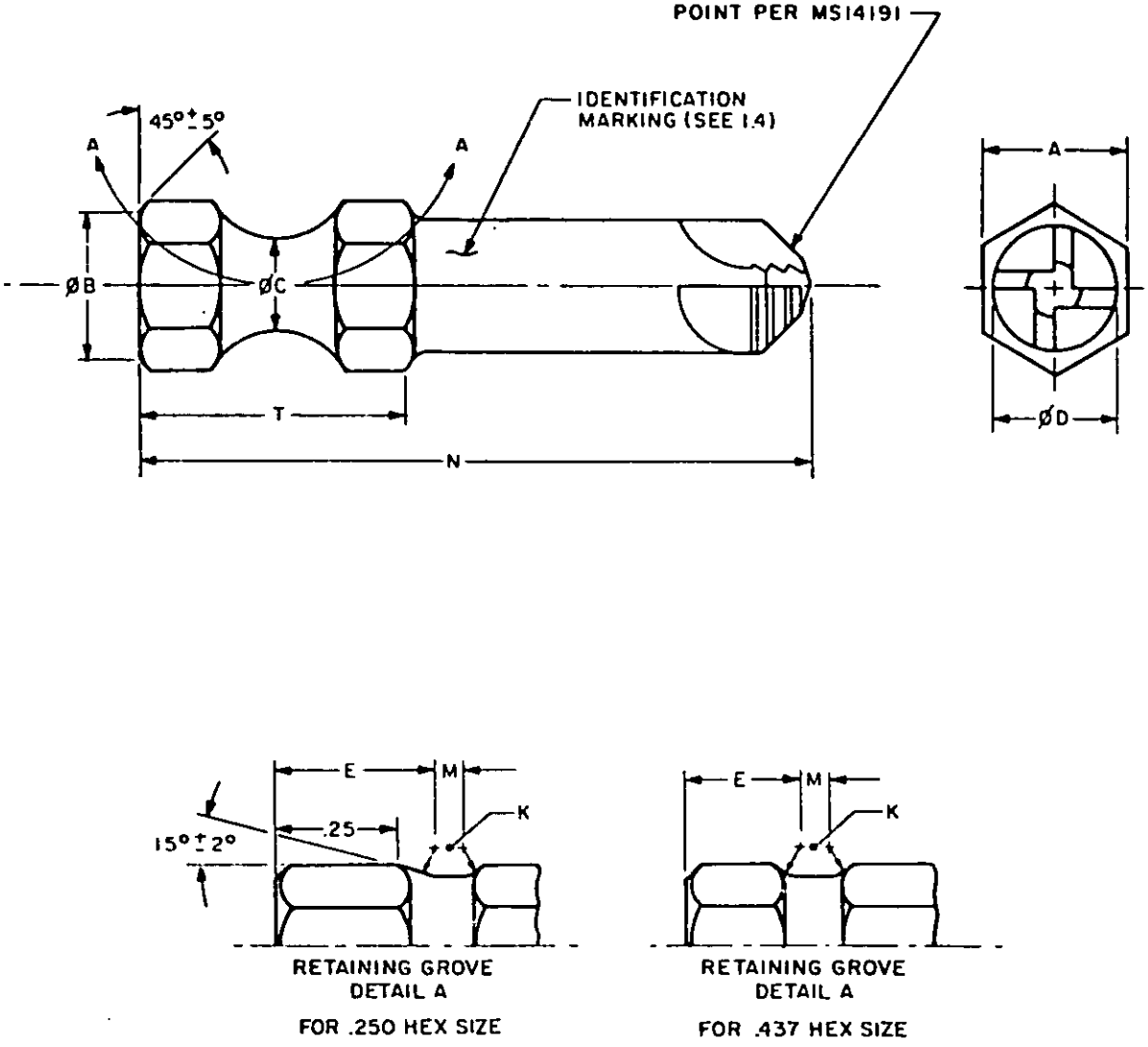
1.6 Minimum ultimate torque loads: In accordance with Table I.

1.7 Torque fatigue loads: In accordance with Table I.

AMSC N/A FSC 5120

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DETAIL A. RETAINING GROOVE.

FIGURE 1. Bit configuration.

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TABLE I. Size and dimensions.

Driver Number	Recess Size	A +.000 -.004	ØB Ref	ØC ±.010	ØD	E ±.020	K RAD	M ±.020	T	Min Ult Torque (Lbf-In.)	Fatigue Torque (Lbf-In.)
-06	6	.250	.203	.188	.188 .172	.344	.094	.062	1.375 0.625	35	16
-08	8				.250 .234					65	30
-03	10									100	45
-04	1/4	.437	.391	.344	.344 .328	.250	.109	.047	1.375 1.000	250	115
-05	5/16				.375 .359					500	200

DIMENSIONS ARE IN INCHES.

TABLE II. Length and dash number.

RECESS AND DRIVER SIZE	N +.12			
	1.94	2.75	3.50	6.00
	DASH NUMBER INDICATING RECESS SIZE AND BIT LENGTH			
6	0619	0627	0635	0660
8	0819	0827	0835	0860
10	319	327	335	360
1/4			435	460
5/16			535	560

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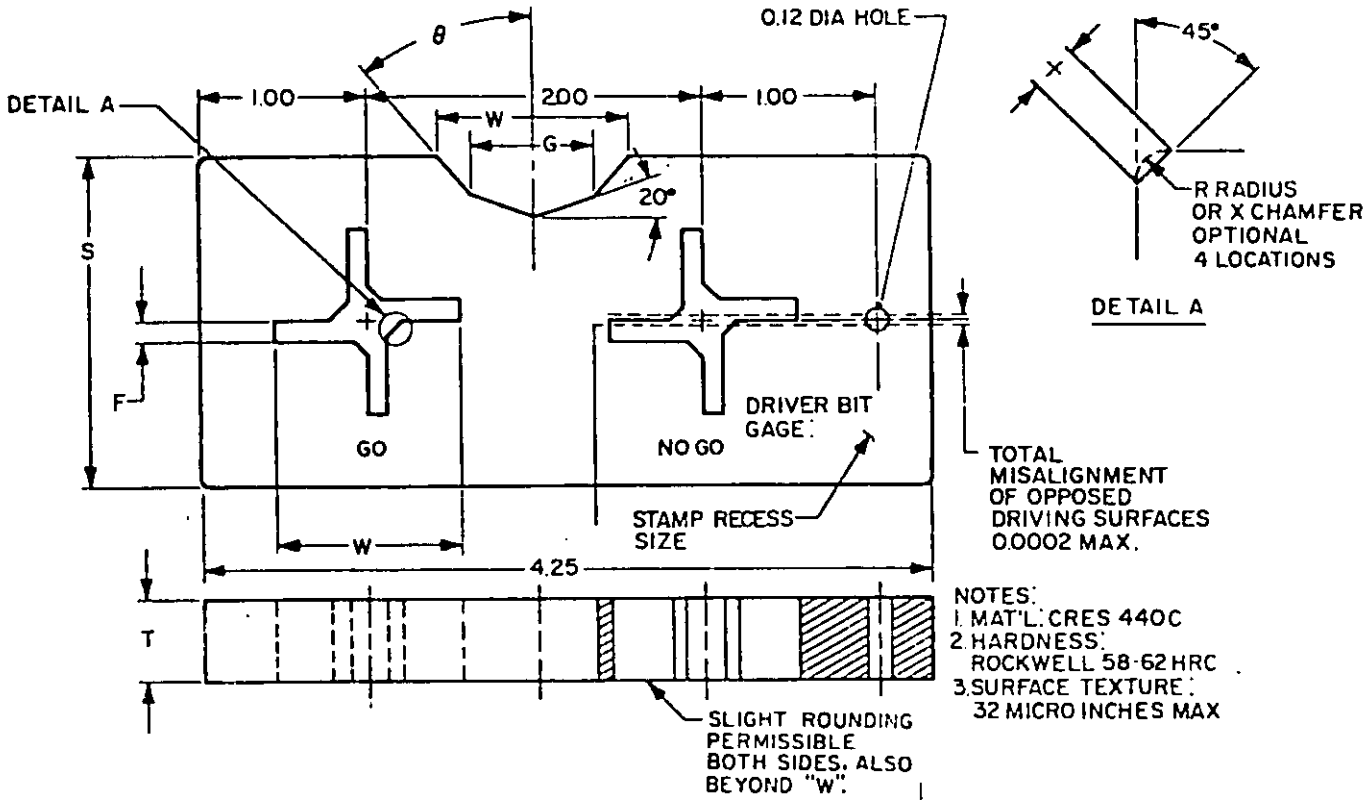


FIGURE 2. Driver bit gage.

TABLE III. Dimensions for offset cruciform driver bit gage.

SIZE	S	GO GAGE DIMENSIONS							NO-GO GAGE DIMENSIONS					
	MIN (WIDTH)	T +.005 (THICK)	W ±.005	G ±.005	F +.0002 -.0000	R +.000 -.001	X +.000 -.001	+1° -0°	W ±.005	G ±.005	F +.0000 -.0002	R +.000 -.001	X +.000 -.001	+1° -0°
6	1.00	.187	.188	.083	.0255	.018	.026	47	.188	.083	.0250	.018	.026	47
8	1.00	.187	.223	.097	.0305	.018	.026	47	.223	.097	.0300	.018	.026	47
10	1.00	.187	.258	.112	.0355	.025	.036	47	.258	.112	.0350	.025	.036	47
1/4	1.00	.187	.340	.148	.0475	.025	.036	48	.340	.148	.0455	.025	.036	48
5/16	1.00	.187	.372	.186	.0605	.031	.044	48	.372	.186	.0585	.031	.044	48

ALL DIMENSIONS IN INCHES.
TOLERANCE: ±0.015 UNLESS OTHERWISE NOTED.

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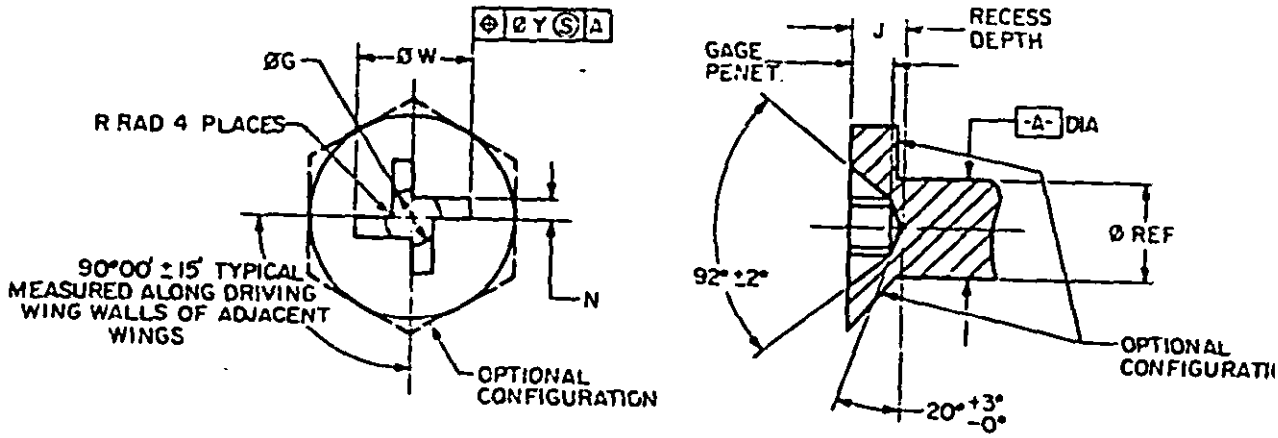


FIGURE 3. Test recess.

TABLE IV. Dimensions for test recess with restricted gage penetration for testing offset cruciform bits.

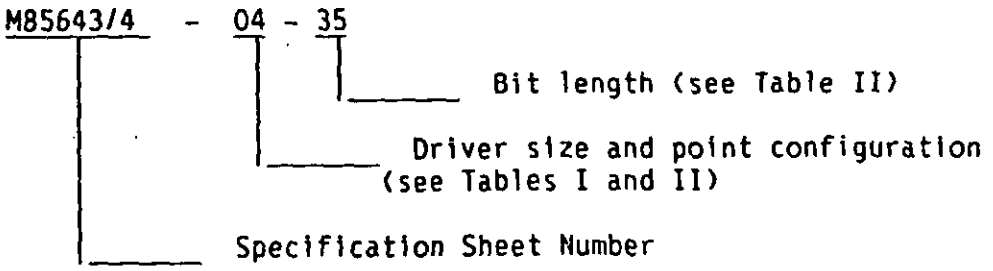
NOMINAL RECESS SIZE	DIMENSION									
	N		G DIA		R RAD		Y DIA	/W WING DIA	J RECESS DEPTH MAX REF	GAGE PENET
	MAX	MIN	MAX	MIN	MAX	MIN	TOL			
6	.029	.026	.082	.078	.030	.016	.012	.182 .172	.066	.0500 .0480
8	.034	.031	.096	.092				.215 .205	.078	.0595 .0575
10	.039	.036	.111	.107	.040	.023		.248 .238	.090	.0685 .0665
1/4	.051	.048	.147	.143			.016	.325 .315	.118	.0890 .0870
5/16	.064	.061	.184	.180	.052	.031		.357 .347	.122	.0860 .0840

NOTES:

1. DIMENSIONS ARE IN INCHES.
2. MATERIAL: UNS K74015 OR UNS T20811 ALLOY STEEL PER AMS 6485 OR AMS 6487.
3. HEAT TREATMENT: 220,000 PSI MINIMUM TENSILE STRENGTH. ROCKWELL 46-49HRC.
4. FINISH: CADMIUM PLATE PER QQ-P-416, TYPE II, CLASS 2.
5. RECESSES TO BE GAGED PER MS14191 EXCEPT GAGE PENETRATION IS LIMITED TO THIS DRAWING.

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1.8 Example of driver part number:



1.9 Apex Machine and Tool Company, Dayton, OH, and Zephyr Manufacturing Company, Inglewood, CA, are the only companies licensed to manufacture ribbed (ACR) Torq-Set bits under United States Patent Number 4,187,892. Patent 4,187,892 expires 12 February 1997. The Government does not have a royalty free license. ACR and Torq-Set are registered trademarks of Phillips Screw Company.

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